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New paleomagnetic results from the Yangzhuang Formation of the Jixian System, North China, and tectonic implications

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Abstract **A paleomagnetic study was carried out on the Yangzhuang Formation (~1350 Ma) of the Jixian System in the North China Block (NCB). Detailed stepwise thermal demagnetization isolated two components. The soft component (component A) was interpreted as a remagnetization in the recent geomagnetic field. The hard component (component B) with higher unblocking-temperatures was carried by hematite. Its site-mean direction is *D/I* = 77.6°/-24.3° (κ = 5.4, α_{95} = 18.3°, *N* = 15 sites) before, and *D/I* = 72.2°/11.5° (κ = 24.6, α_{95} = 7.9°) after tilt correction. It passes a fold test (Mesozoic folding) at 99% confidence level and reversal test at 95% confidence level. The corresponding pole locates at 17.3°N, 214.5°E (*dp* = 4.1°, *dm* = 8.0°). A best fit of paleomagnetic poles from the NCB, Baltica, Siberia and the apparent polar wander path for Laurentia suggests a long-lived connection between these paleo-continents between ~1800 and 1350 Ma.**

Keywords: North China Block, Yangzhuang Formation, paleomagnetism, Mesoproterozoic, Rodinia, Columbia.

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Precambrian continental reconstructions between the North China Block (NCB) and other continents have been debated over a decade[1−6], partly due to the sparsity of reliable paleomagnetic poles from these continents for the Precambrian time. The Precambrian successions (ca. 1800−800 Ma) in the Jixian County in North China are well exposed with several reddish carbonate intervals that are suitable for paleomagnetic study. Although a number of paleomagnetic investigations have been carried out previously on these successions[7−9], the reliability of most earlier results is questionable due to one or more of the following reasons: unrecognized Mesozoic magnetic overprint, insufficient sampling, and lack of modern demagnetization and field test. In this paper, we report a stable magnetic component from purple muddy dolomites in the Yangzhuang Formation (ca. 1350 Ma) that is interpreted as a primary remanence, and discuss possible links between the NCB and other continents.

1 Regional geology and sampling

In the NCB, the Mesoproterozoic to Neoproterozoic stratotype section is located at northern Jixian County, ca. 100 km east of Beijing (Fig. 1(a)). The strata in this area consist of 10 km-thick unmetamorphosed section of clastic and carbonate rocks and are divided into three systems: the Changchengian (ca. 1800−1400 Ma), the Jianxianian (ca. 1400−1000 Ma) and the Qingbaikouan (ca. 1000−800 Ma). The Yangzhuang Formation at the bottom of the Jixian system is composed of purplish-red and greyish white muddy dolomite[10]. It disconformably overlies the Gaoyuzhuang Formation and is overlain conformably by the Wumishan Formation. The only isotopic age (1317 ± 22 Ma) is an Ar-Ar age of gas enclosed in chert from the upper Yangzhuang Formation[11]. Together with age constraints from the Gaoyuzhuang and Wumishan Formations, Wang et al.[11] gave estimated ages of 1380 ± 24 Ma and 1310 ± 20 Ma for the lower and upper boundaries of the Yangzhuang Formation, respectively. Samples for this study were collected from Member 2 and lower Member 3 of the Yangzhuang Formation, with an estimated age of ca. 1350 Ma.

A total of 212 core samples from 23 sites were collected in northern Jixian County (Fig. 1(b)) from purple muddy dolomites in the Yangzhuang Formation using a portable gasoline-powered drill and oriented with a magnetic compass and a sun compass whenever possible.

2 Analytical methods

The samples were cut into 2.2 cm-long standard paleomagnetic specimens using a diamond saw. The magnetic remanence was measured using either a JR-6A spinner magnetometer in the Paleomagnetic Laboratory of China University of Geosciences (Beijing), or a 2G-755R cryogenic magnetometer in the Paleomagnetic Laboratory of the Institute of Geomechanics in Beijing. All samples were subjected to stepwise thermal demagnetization using ASC-TD48 ovens with an internal residual field below than 10 nT. Magnetometers and ovens in both laboratories are hosted inside large Helmholtz coils. During the experimental procedure, samples were further protected from magnetic contamination using μ-metal boxes. Stepwise thermal demagnetization was carried out in steps of 50−100℃ at temperatures < 500℃ and 5−30℃ at temperatures >500℃ until the directional behavior became erratic. Principal component analysis[12] or stable endpoints methods were used for extracting remanence directions from the majority of the samples. Statistics on the distribution of directions were calculated using Fisher

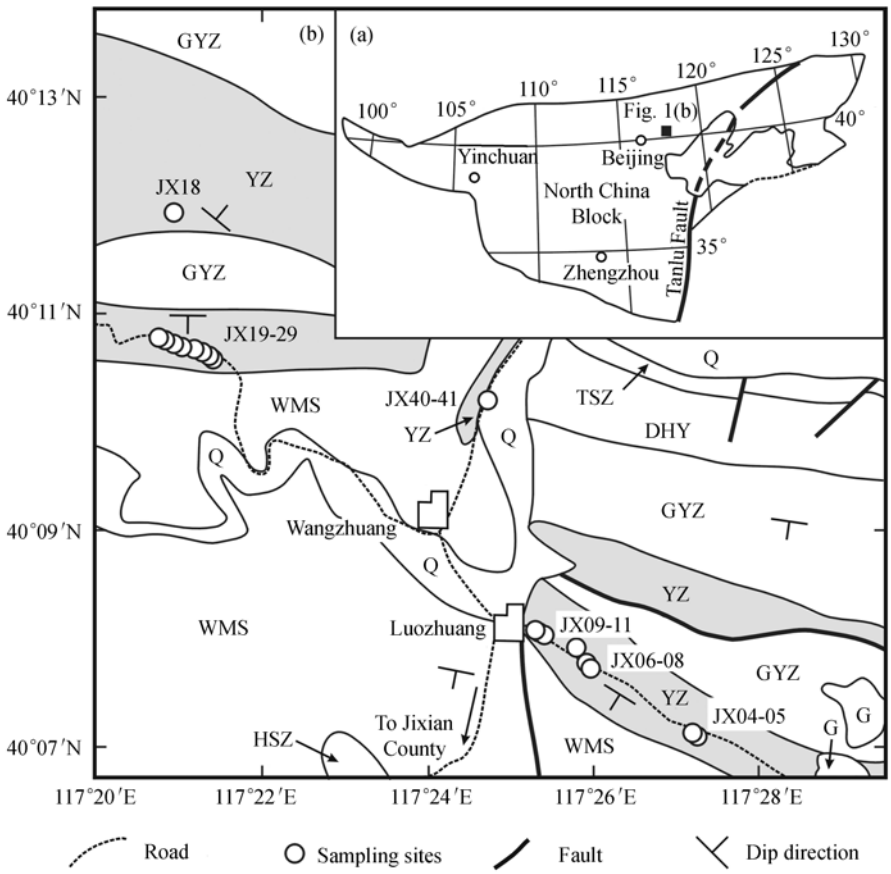


Fig. 1. (a) Tectonic outline of the North China Block; (b) simplified geological map of the sampling area in northern Jixian County. Shadowed areas show the distribution of the Yangzhuang Formation. Locations of sample sites are showed as open circles. TSZ, Tuanshanzi Formation; DHY, Dahongyu Formation; GYZ, Gaoyuzhuang Formation; YZ, Yangzhuang Formation; WMS, Wumishan Formation; HSZ, Hongshuizhuang Formation; G, Intrusions; Q, Quaternary.

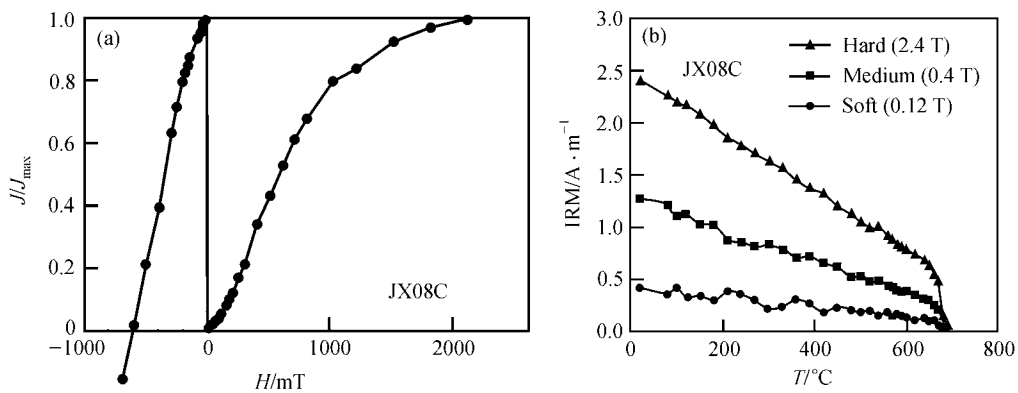


Fig. 2. (a) Normalized acquisition curve and opposite field demagnetization of isothermal remanent magnetization; (b) stepwise thermal demagnetization of three-component IRM following Lowrie[15].

statistics[13]. Site-mean directions for samples from three sites whose remanence directions became erratic after 670℃ were obtained using great-circle intersections in combination with directly observed directions[14]

3 Palaeomagnetic results

Rock magnetic studies were carried out to identify the magnetic minerals in representative samples. The IRM acquisition curve of a reddish dolomite sample shows that magnetic saturation is not achieved at a field of ca. 2.4 T. Remanence coercivity of the same sample is about 0.6 T (Fig. 2(a)). Three-component IRM thermal demagnetization (Lowrie test)[15] indicates that the unblocking tem-

perature is around 680℃ for such samples (Fig. 2(b)). Both experiments indicate hematite as the dominant magnetic mineral.

The natural remanent magnetization (NRM) intensities of the purple muddy dolomite samples are in the range of 0.5—5 mA/m. After removal of a stray viscous remanent magnetization under 200℃, two components were found in most samples (Fig. 3). A soft component A is often isolated at the temperature range of 100—670℃. The mean direction of component A is close to the present geomagnetic field direction and the directions of component A become scattered after tilt correction (Fig. 4(a)). This suggests that component A is an overprint magnetization acquired during chemical weathering in recent geomagnetic field.

A very different high temperature component B with dual polarities is found in most samples toward the end of the demagnetization sequence, typically at temperatures above 670℃ (Fig. 3). Component B directs either to east-northeast with shallow downward inclinations, or to west-southwest with shallow upward inclinations, both in stratigraphic coordinates (Fig. 4(b)). Site-mean direction of component B is $D/I = 77.6°/-24.3°$ ($\kappa = 5.4$, $\alpha_{95} = 18.3°$, $N = 15$ sites) before, and $D/I = 72.2°/11.5°$ ($\kappa = 24.6$, $\alpha_{95} = 7.9°$) after tilt correction. It passes the fold test of McFadden[16] at 99% confidence level and reversal test of McFadden and McElhinny[17] at 95% confidence level ($\gamma = 14.3° < \gamma_{critical} = 14.7°$, classification C) (Table 1, Fig. 4(b)). The fold was formed during the Middle to Late Jurassic[10].

Fig. 5 shows that the geomagnetic field reversed several times during the deposition of Members 2 and 3 of the Yangzhuang Formation. The corresponding paleomagnetic pole for component B locates at 17.3°N, 214.5°E (dp/dm = 4.1°/8.0°) and is termed pole "YZ" here. Our new result is quite different from that of Zhang and Zhang[7] and Lin[8], but is consistent with that of Zhang et al.[9]. However, in this paper, we for the first time present detailed demagnetization information, a positive fold test, a reversal test, a magnetostratigraphic column, and structural information.

Pole YZ is also close to the Silurian-Devonian pole of Zhao et al (1993)[18] from the Hexi Corridor in western North China. However, we suggest that pole YZ is unlikely a Silurian-Devonian overprint because (1) a later

Table 1 Site mean directions and virtual geomagnetic poles (VGPs) for component B from the Yangzhuang Formation (40.1°N, 117.4°E)

Site	Bedding	n/n_o	Direction (*in situ*)				Direction (tilt-corrected)		VGP			
	dip/dip azimuth/(°)		D/(°)	I/(°)	κ	α_{95}/(°)	D/(°)	I/(°)	Lat./(°N)	Long./(°E)	dp/(°)	dm/(°)
JX04	46/208.5	8/9	243.5	35.4	11.7	16.9	236.5	–4.6	–26.6	48.7	8.5	16.9
JX06	55/247	4/7	292.2	53.6	56.8	12.3	272.1	6.8	3.8	28.7	6.2	12.4
JX07	36/224	11/13	243.6	21.4	17.9	11.1	242.7	–12.8	–24.9	40.6	5.8	11.3
JX08	36/210	6/6	68.5	–18.5	121.6	6.1	66.8	9.8	20.8	218.9	3.1	6.2
JX09	89/238	8/8	293.8	59.8	282.6	3.3	256.1	–12.8	–14.8	31.4	1.7	3.4
JX10	83/236	12/12	304.4	49.4	70.6	5.2	272.1	–7.1	–0.7	23.3	2.6	5.2
JX11	83/236	13/14	297.9	56.0	103.2	4.1	264.5	–8.2	–6.9	27.8	2.1	4.1
JX18	41/043	3/3	68.6	68.7	34.9	21.2	53.2	29.0	37.9	219.6	12.9	23.4
JX19	67/178	8/8	68.0	–2.7	58.9	7.3	77.2	16.9	15.3	209.0	3.9	7.5
JX20 a)	62/182	6/15	54.4	–2.1	25.7	15.9	65.4	24.7	27.0	213.2	9.1	17.1
JX22	65/186	4/8	65.7	0.1	131.2	8.1	82.1	27.2	15.2	201.2	4.8	8.8
JX23a)	65/186	12/12	68.5	–10.5	27.5	8.7	73.7	19.6	18.9	210.1	4.8	9.1
JX27 a)	65/187	5/7	85.4	–12.8	49.9	13.6	81.2	4.6	8.2	211.3	6.8	13.6
JX28	59/186	6/6	258.4	32.9	29.9	12.5	239.3	3.6	–21.7	49.7	6.3	12.5
JX29	62/183	6/6	253.5	28.4	90.0	7.1	239.1	–2.1	–23.9	47.7	3.6	7.1
"Normal" polarity		7 sites	68.3	1.2			71.6	19.1	20.4	211.7	5.4	10.3
"Reversed" polarity		8 sites	268.5	44.7			252.7	–4.8	–14.7	36.9	5.9	11.7
Mean direction†		15 sites	77.6	–24.3	5.4	18.3						
		15 sites			24.6	7.9	72.2	11.5	17.3	214.5	4.1	8.0

† McFadden fold test[16] (definition 2) is positive at 99% confidence level: $N = 15$, SCOS 95% = 4.510, SCOS 99% = 6.305; *In situ:* SCOS2=8.767; Tilt-corrected: SCOS2=1.629.
McFadden and McElhinny[17] reversal test is positive at 95% confidence level: $\gamma = 14.3° < \gamma_{critical} = 14.7°$, classification C.

a) Results are calculated from intersections of the remagnetization great circles and line fitting[14]. The assigned polarities are arbitrary, thus marked with " ". n/n_o: number of specimens used in the calculation/number of specimens demagnetized; $D(I)$: declination (inclination) of a direction; κ: Fisher's precision parameter; α_{95}: half-angle of the 95% confidence cone for a mean direction. VGPs are calculated from tilt corrected directions. Lat.(°N)/Long.(°E): north latitude/east longitude of VGP. dp/dm: semi-axes of the oval 95% confidence for a VGP.

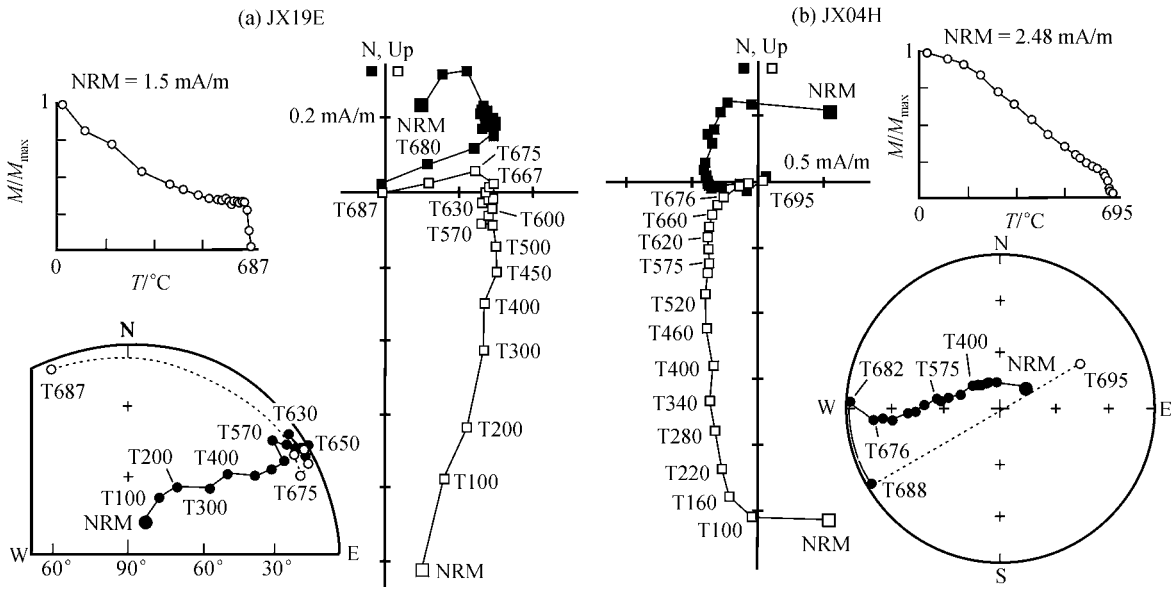


Fig. 3. Examples of stepwise thermal demagnetization data for samples from the Yangzhuang Formation (orthogonal vector diagrams, equal-area stereographic projections and normalized demagnetization). All directions are *in situ*. Solid (open) squares in the orthogonal vector diagrams represent horizontal (vertical) projections. Solid (open) symbols on the stereographic projections denote downward (upward) directions. NRM = natural remanent magnetization; T100 = 100℃.

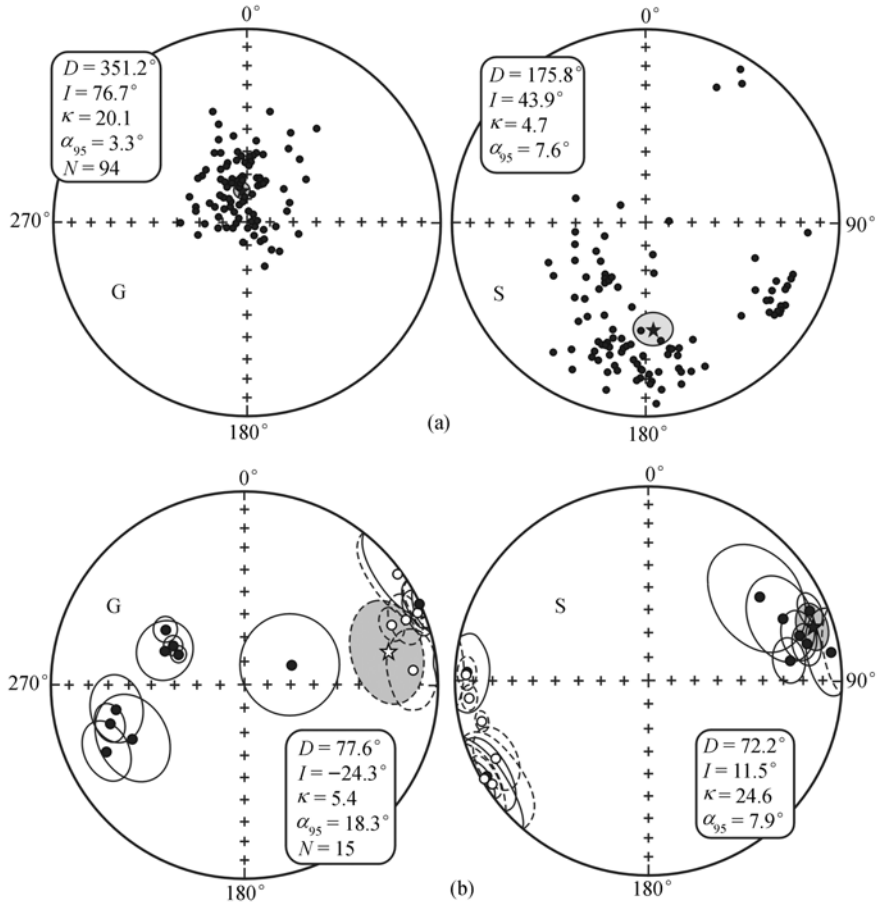


Fig. 4. Equal-area stereographic projections of component A (a) and component B (b) from the Yangzhuang Formation. G/S represents geographic/stratigraphic coordinates. Solid (open) symbols on the stereographic projections denote downward (upward) directions. Stars with shaded 95% confidence circles represent mean directions.

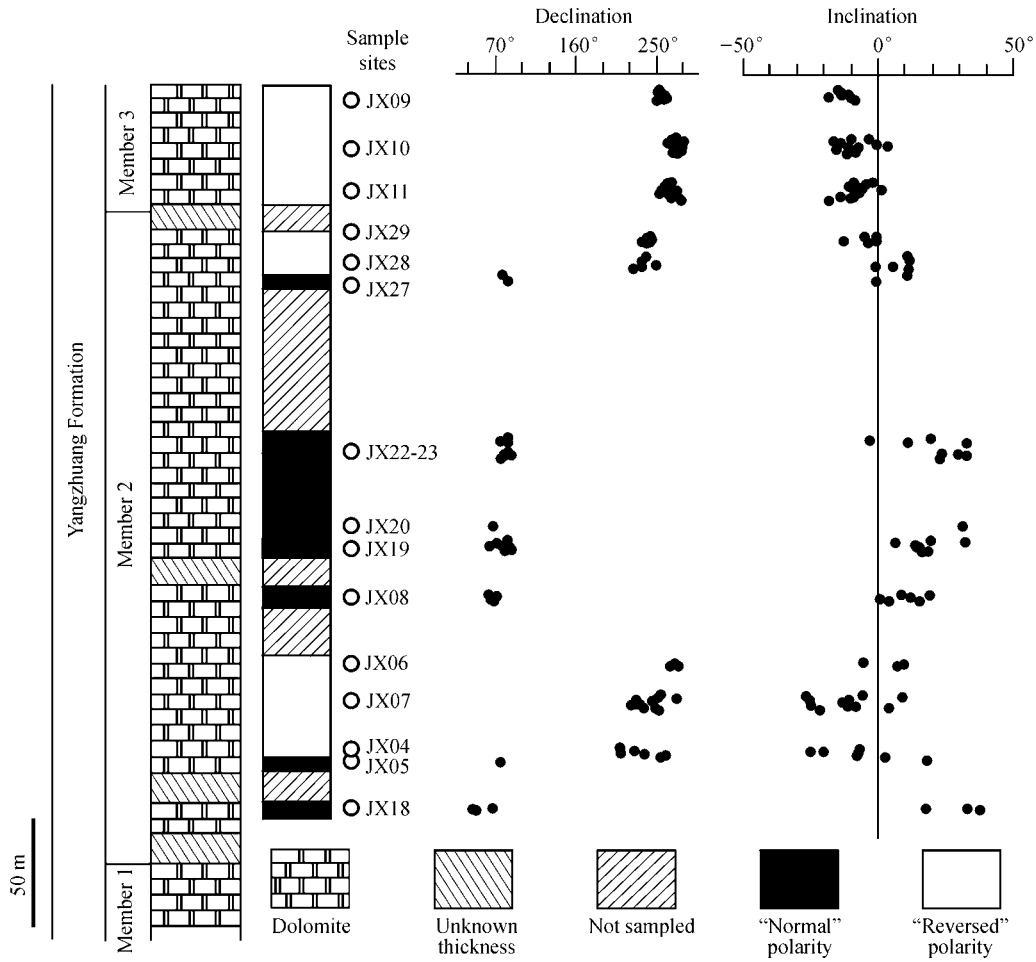


Fig. 5. Simplified lithological column and magnetostratigraphy of the Yangzhuang Formation. The assigned polarities are arbitrary. The northeast, positive directions are assumed as the "Normal" polarity.

Table 2 Selected paleomagnetic poles used for reconstructions[a)]

Pole abbreviation	Rock units	Lat./(°N)	Long./(°E)	A_{95}/(°)	Age/Ma	Reference
North China Block (NCB)						
TH	Taihang dyke swarm	36.0	247.0	2.8	1769.1±2.5	[30]
YZ	Yangzhuang Formation	17.3	214.5	5.7	~1350Ma	This study
Laurentia						
MD	Molson dykes	15.4	263.5	4.0	~1820－1720	[30]
MA	Michikamau anorthosite pluton	–2.0	218.0	5.0	~1460	[32]
HL	Harp Lake complex	2.0	206.0	4.0	~1450	[32]
LA	Laramie complex and Sherman granite	–7.0	215.0	4.0	~1434	[32]
MQ	Mistastin complex	–1.0	201.0	8.0	~1420	[32]
NA	Nain anorthosite	12.0	210.0	3.0	1320－1290	[32]
MAD	Mackenzie dolerite dykes	4.0	190.0	5.0	1267±2	[32]
Baltica						
SF	Shoksha Formation, Vepsian Group	39.7	221.1	4.0	~1780	[33]
SQ	Subjotnian quartz porphyry dykes	29.0	177.0	6.0	~1630	[32]
AD	Åland dolerite dykes	28.0	188.0	9.0	1577±12	[32]
AQ	Åland quartz porphyry dykes	12.0	182.0	7.0	1571±20; 1571±9	[32]
MJD	Mean Jotnian dolerite intrusions	4.0	158.0	4.0	~1265	[32]
Siberia						
CD	Chieress dyke	4	258	6.6	1384±2	[34]

a) Lat.(°N)/Long.(°E): north latitude / east longitude of VGP; A_{95} is 95% confidence circle of the pole.

report[19] indicates that local rotations could not be ruled out for Zhao et al.'s[18] sampling location (Fig. 9 in ref. [19]); (2) no widespread Silurian-Devonian remagnetization has been reported from the NCB. Pole YZ is located far from all other reported poles from the younger strata in the NCB[4,19−23]. It satisfies all 7 reliability criteria proposed by Van der Voo[24] and we thus regard it as a "key pole" for the NCB.

4 Discussions

The concept of the Paleo- to Mesoproterozoic supercontinent, called Columbia, has recently been proposed: its assembling occurred during 2.1−1.8 Ga, its breakup began at 1.6−1.4 Ga, and its final fragmentation occurred at 1.3−1.2 Ga[5, 25−28]. Pieces of continents broken away from Columbia combined to form Rodinia. However, both the very existence of Columbia and its configuration are being debated[29]. Our new high quality paleomagnetic pole from the NCB may help to constrain the role of the NCB in the assembly and breakup of the proposed supercontinent Columbia.

We list in Table 2 all paleomagnetic poles used for our reconstruction. For the NCB, apart from the ca. 1350 Ma YZ pole reported here, there is also a precisely dated pole of 1769.1 ± 2.5 Ma with positive contact test from dyke swarms in the Taihang region, Shanxi Province[30] (Table 2). For Laurentia, pole MD is a remagnetization pole assigned with a 1.82−1.72 Ga age, derived from the 1.88 Ga Molson dykes adjacent to the Trans-Hudson orogen[30] (Table 2). Pole NA for Laurentia was from the Nain anorthosite complex (1320−1290 Ma) that plots within the 1460−1265 Ma segment of the Laurentia APWP[31,32]. This second pole could be a primary pole although there is no field test for it[32]. For Baltica, pole SF was from the ca. 1780 Ma red sandstones in the Vazhinka River section, Karelia, where magnetic polarity zonations were reported and the stable remanence passes the reversal test. This pole has been regarded as a key pole[33] (Table 2). There is no key pole for Siberia for the 1400−1300 Ma interval. The only precisely dated paleomagnetic pole was a VGP from an east-southeasterly trending dyke of the east-southeast trending 1384 ± 2 Ma Chieress swarm in the Anabar Shield[34] (Table 2). Ernst et al.[34] regarded this pole as of primary origin on the basis that Archaean gneisses and schists from the same area have a different remanence direction and the dykes are almost non-metamorphosed.

Our reconstruction in Fig. 6 is made through matching the paleomagnetic poles of NCB, Baltica and Siberia with the APWP of Laurentia for the ca. 1800−1260 Ma interval: northern NCB is placed against northern Siberia whose southern margin is adjacent to northwestern Laurentia; northwestern Baltica is placed against Greenland.

In this configuration, pole SF from Baltica, TH from the NCB and MD from Laurentia, all of ca. 1800−1750 Ma ages, plot next to each other. Pole YZ from NCB, NA from Laurentia, and CD from Siberia, all of ca. 1380−1290 Ma, overlap each other. Pole MJD from Baltica and MAD from Laurentia are both of ca. 1265 Ma, and their positions overlap. It is thus plausible that the NCB, Laurentia, Baltica and Siberia shared a long-lived connection from ca. 1800 Ma till 1350 Ma or later. We note that due to the polarity uncertainties for the selected poles, this reconstruction is non-unique.

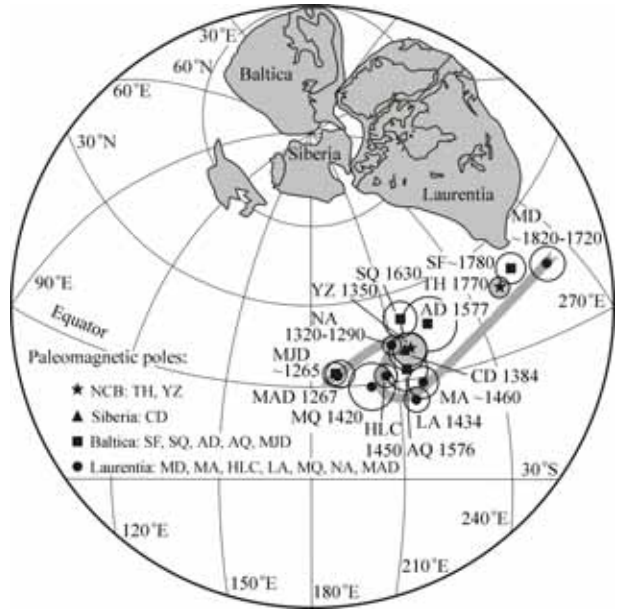


Fig. 6. A possible paleogeographic reconstruction between Laurentia, Baltica, Siberia and the North China Block (NCB) for the time interval of ca. 1800−1350 Ma. Rotated according to present North America coordinates. Paleopoles from different continents are shown with different symbols with numbers indicating their ages in Ma. Paleopoles, corresponding abbreviations, ages and references are given in Table 2. Euler rotation parameters (+/- represent counter clockwise and clockwise rotations): NCB to Laurentia (10.81°N, 196.44°E, –24.04°), Baltica to Laurentia (51.80°N, 353.75°E, 35.89°), Siberia to Laurentia (77.14°N, 113.19°E, 138.73°).

Our reconstruction is consistent with geological constraints. Karlstrom et al.[35] proposed that a series of subparallel accretionary orogens of 1.8−1.0 Ga at the southern margin of Laurentia may extend to Baltica. Based on the comparisons of Archaean and Paleoproterozoic crustal provinces, Laurentia and Siberia are proposed to have been together at that time interval[36,37], or even until early Cambrian[38]. The long-lived links between the NCB, Laurentia, Baltica and Siberia during ca. 1800−1350 Ma as proposed here are consistent with those proposed using geological and paleontological evidence[1−3, 6]. The connection between the NCB and Laurentia is particularly similar to that proposed by Li et al.[3]. Unlike the suggestion of Rogers and Santosh[27] that the NCB was separated from Baltica between 1.7−1.6 Ga, our work indicates that the NCB may have been connected to Laurentia, Baltica

and Siberia till ca. 1350 Ma. In this case, the Mesoproterozoic Zhaertai-Bayan Obo rift[5,39,40], which has poor age constraint, might have resulted from rifting between the NCB and Siberia.

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